

SECRETARY ISLAND, New Zealand

WMO: 938000

Lat: 45.221S

Lon: 166.885E

Elev: 62

StdP: 14.66

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.9	39.6	28.1	22.1	43.6	30.5	24.7	43.7	33.2	52.7	30.5	52.1	6.2	150	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.1	70.2	61.1	68.1	60.6	66.3	60.0	64.1	67.1	62.7	66.1	61.3	64.6	9.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.0	86.2	65.5	61.2	80.9	64.1	59.8	76.9	63.1	29.3	67.2	28.3	66.1	27.4	64.9	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
31.9	28.5	25.4	DB	35.6	74.9	1.3	2.6	34.6	76.8	33.9	78.4	33.1	79.9	32.2	81.8	
			WB	32.9	65.2	1.1	2.2	32.1	66.7	31.5	68.0	30.9	69.3	30.1	70.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.3	58.6	59.2	57.7	55.0	52.2	49.3	48.4	49.1	50.2	51.0	52.8	56.2
	DBStd	5.13	4.00	3.75	3.74	3.47	3.78	3.30	3.21	3.04	3.21	3.30	3.61	3.87
	HDD50	273	0	0	1	4	20	52	68	52	36	26	12	2
	HDD65	4285	202	165	228	299	395	472	514	494	445	433	365	273
	CDD50	1471	267	257	239	155	90	30	19	23	41	58	98	194
	CDD65	7	4	2	1	0	0	0	0	0	0	0	0	0
	CDH74	8	4	2	1	0	0	0	0	0	0	0	0	1
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	10.1	10.5	10.2	10.1	9.7	9.7	8.5	9.5	9.4	10.8	10.8	10.7	10.8
Precipitation	PrecAvg	149.9	14.1	11.8	12.8	12.1	14.3	11.1	9.9	10.2	11.7	14.1	13.7	14.3
	PrecMax	191.0	24.6	25.2	30.4	28.1	26.5	18.1	15.2	17.5	16.9	24.1	27.3	23.2
	PrecMin	115.7	3.0	5.1	4.1	4.8	5.9	4.4	4.1	4.5	3.0	7.8	5.3	5.7
	PrecStd	16.4	5.1	4.6	5.7	5.1	4.4	3.6	2.9	3.0	3.7	4.5	5.3	4.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.5	72.6	70.8	66.6	63.7	58.9	58.2	59.6	61.7	64.5	67.6	71.4
		MCWB	62.6	62.0	60.1	56.7	56.4	52.8	50.6	50.4	53.1	54.3	57.7	61.0
	2%	DB	70.1	69.6	67.6	63.9	60.8	57.0	56.0	56.6	58.9	60.7	63.6	67.8
		MCWB	61.7	61.4	60.5	56.8	55.8	52.5	50.4	50.6	52.0	53.5	55.6	59.8
	5%	DB	67.8	67.5	65.7	62.1	59.1	55.7	54.6	55.2	56.9	58.3	60.8	65.0
		MCWB	61.0	60.9	60.3	56.5	55.0	52.1	50.2	50.2	51.2	52.4	54.7	58.7
	10%	DB	65.4	65.6	64.0	60.4	57.8	54.4	53.3	53.7	55.2	56.4	58.7	62.7
		MCWB	59.8	60.6	59.4	56.1	54.2	51.2	49.6	49.7	50.8	52.2	54.1	57.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.3	65.6	65.2	60.7	59.7	56.1	54.5	53.9	55.5	57.0	60.1	63.4
		MCDB	69.0	68.6	66.6	62.6	61.2	57.0	55.9	55.5	58.9	60.3	63.9	67.8
	2%	WB	63.8	63.9	63.4	59.2	57.7	54.4	52.9	52.6	53.8	55.4	57.6	61.5
		MCDB	67.6	67.0	65.5	61.4	59.4	55.8	54.5	54.7	56.2	57.6	60.7	65.0
	5%	WB	62.1	62.5	61.5	58.1	56.3	53.0	51.6	51.4	52.8	54.2	56.3	60.1
		MCDB	65.9	65.8	64.2	60.5	58.1	54.8	53.4	53.9	55.2	56.7	59.3	63.4
	10%	WB	60.6	61.1	60.1	56.9	54.9	51.7	50.4	50.4	51.8	53.1	55.0	58.7
		MCDB	64.0	64.6	63.1	59.6	57.1	53.8	52.5	53.1	54.5	55.7	58.0	62.3
Mean Daily Temperature Range	5% DB	MDBR	10.1	10.1	9.5	9.2	8.1	8.1	8.2	8.7	8.7	9.1	9.5	10.0
		MCDBR	14.3	13.4	12.8	11.4	9.3	9.3	9.7	10.6	11.5	12.4	13.6	13.7
		MCWBR	8.7	8.5	8.0	7.7	6.9	7.1	7.1	7.3	7.9	8.1	8.8	8.6
	5% WB	MCDBR	10.8	10.4	8.9	8.2	7.1	7.0	6.3	8.1	8.8	8.8	9.8	10.8
		MCWBR	7.5	7.6	6.9	7.0	6.5	6.7	6.0	6.6	7.1	6.9	7.3	7.4
Clear-Sky Solar Irradiance	taub	0.351	0.340	0.325	0.317	0.311	0.305	0.299	0.312	0.348	0.342	0.339	0.336	
	taud	2.528	2.575	2.625	2.628	2.607	2.577	2.608	2.592	2.470	2.506	2.532	2.574	
	Ebn at Noon	308	302	292	270	245	233	247	267	278	298	310	315	
	Edh at Noon	34	30	26	22	19	17	18	22	30	32	34	33	
All-Sky Solar Radiation	RadAvg	1741	1508	1117	733	430	341	403	631	941	1316	1654	1821	
	RadStd	148	113	83	63	39	36	40	40	84	103	168	134	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.56	+0.58	N/A	N/A	N/A	N/A	-54	-225	+151	N/A
Regional (3 neighbors)	N/A	N/A	+1.39	N/A	N/A	N/A	-85	-202	N/A	N/A

Nomenclature: See separate page